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NAD boosting mediated by CD38 inhibition drives reversal of a pathological vicious cycle of intracrine activity and inflammation in eyelid meibomian gland dysfunction

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eLife Assessment

The authors provide **valuable** data linking NAD⁺ dependent HSD3b6 gene expression in the eyelid to a vicious cycle involving decreased steroidogenesis and AR signaling, pro-inflammatory cytokine release, inflammation, CD38 activation, and further NAD⁺ decline, which induces meibomian gland atrophy leading to dry eye disease. Overall, the presented work provides evidence for the pathologic relationship between a pro-inflammatory environment, intracrine activity, and the NAD⁺ cofactor. However, the current study does not clearly establish the proposed intracrine mechanism and may largely reflect systemic hormonal effects resulting from the global Had3b6 knockout, leading to an **incomplete** narrative.

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Abstract

Vicious cycles—reciprocal cause-and-effect loops that progressively intensify dysfunction—are implicated in many diseases. Here, we examined whether “intracrine” steroidogenesis, a tissue-local mode of hormone production, participates in a pathological vicious cycle in eyelid meibomian gland dysfunction (MGD), the most common cause of evaporative dry eye disease with incidence increasing with age. We show that loss of meibomian intracrine androgen production triggers inflammatory remodeling of the eyelid tarsal plate. Conversely, inflammation suppresses nicotinamide adenine dinucleotide (NAD)-dependent 3 β -hydroxysteroid dehydrogenase activity in the meibomian gland, thereby further weakening intracrine steroidogenesis. Mechanistically, we identified inflammation-associated accumulation of the NAD-degrading enzyme CD38 as a key down-regulator for the NAD-dependent intracrine activity. Accordingly, pharmacological inhibition of CD38 (using 78c) in aged mice, restores meibomian cellular NAD levels, rescues intracrine activity, suppresses inflammatory signatures, and promotes recovery of gland size. Thus, our data suggest that boosting NAD availability not only ameliorates meibomian gland dysfunction but also induces a reciprocal “virtuous” cycle, in which suppression of inflammation and restoration of intracrine activity mutually reinforce each other. These findings constitute a previously unknown example in which the reversal of a pathological vicious cycle into a beneficial virtuous cycle underlies the therapeutic effects of NAD boosting and may help combat age-associated meibomian gland disorder.

Introduction

A vicious cycle is a sequence of reciprocal cause and effect in which two or more elements intensify and exacerbate each other, progressively driving further deterioration of the system. Such self-reinforcing processes are thought to underlie a wide range of diseases, including neurodegenerative disorders^{1,2}, metabolic diseases^{3,4}, and cardiovascular dysfunction^{5,6}. In the context of aging, however, how vicious cycles are initiated and sustained, and how they contribute to disease progression, remains poorly understood. A logical and mechanistic understanding of these vicious cycle processes is of critical importance, as it may provide a basic framework for developing therapeutic strategies aimed at alleviating disease phenotype.

In the present study, we investigated the contribution of “intracrine” activity as a component of the potential vicious cycle in pathology. Particularly we studied age-related tissue disorder resulting from deficient local steroid hormone (intracrine hormone) synthesis. In general, hormones are produced from endocrine organs and delivered to their target tissues. However, for steroids, non-endocrine tissues or cells also participate in hormone production via an intracrine system. In this system, hormones are produced in the tissues in which they exert their effects without being released into circulation^{7–9}. The concept of intracrinology was established in the 1990s^{7–9}, yet its potential participation in pathological vicious-cycle conditions has remained unexplored. We recently reported that attenuated intracrine activity has a causal contribution to the pathogenesis of eyelid meibomian gland dysfunction (MGD)^{10,11}, a common ocular surface disorder whose incidence increases with age^{12,13}. The alteration of intracrine activity has also been described for several extra-gonadal tissues such as uterine endometrium¹⁴ and skin sebaceous gland¹⁵ as well as several cancerous tissues including those of the prostate¹⁶, breast¹⁷ and bone¹⁸. Among those, in our current study, we investigated the possibility of meibomian gland intracrine activity as a component of a disease vicious cycle in the eyelids.

Meibomian glands are modified sebaceous glands located in the tarsal plates of upper and lower eyelids and are responsible for the secretion of meibum (lipid) to the ocular surface to prevent tear evaporation (dry eye). The meibomian gland cells undergo a continuous cell proliferation (and subsequent differentiation) to produce meibum as they employ a holocrine system in which cells themselves die into the meibum after lipid synthesis and accumulation. In this process, androgens are produced inside the meibomian gland cells (more specifically, in the meibomian basal acinar cells, or precursor cells) and act to maintain active cell proliferation (and lipid generation) of these cells^{10,19}. However, with aging, meibomian gland cells undergo reduced cell proliferation, leading to the atrophy of the tissue, which is a common feature of aging in both humans^{20,21} and mice^{22,23}. Clinically, the loss of the meibomian glands results in reduced tear film lipid, unstable tear film, increased aqueous tear evaporation²⁴, and increased tear film osmolarity²⁵, leading to ocular surface changes and blepharitis^{26,27}. Particularly, it is recognized that meibomian gland dysfunction is the most common cause of evaporative dry eye disease in clinical practice^{12,13}. Although evidence suggests that locally generated androgens act within the meibomian gland^{10,28}, a systematic understanding of how altered intracrine activity in the meibomian gland relates to other unknown pathological features in the eyelid tarsal plate is still lacking.

The enzyme 3 β -hydroxysteroid dehydrogenase/isomerase (3 β -HSD) is essential for the biosynthesis of steroid hormones, including androgen produced by the meibomian gland cells²⁹. In the present study, we found that reduced meibomian gland intracrine activity causes tissue inflammation in the tarsal plate, which in turn further attenuates meibomian intracrine activity via 3 β -HSD. Of note, 3 β -HSD is an enzyme whose rate of activity is influenced by the amount of nicotinamide adenine dinucleotide (NAD)^{10,30}—a cofactor required for the dehydrogenase activity of 3 β -HSD. Inflammation in the tarsal plate was found to involve increased expression of CD38—an NAD/NMN-consuming enzyme—which diminishes the availability of NAD in the meibomian gland. Our study was performed to understand this potential vicious cycle as it may provide a mechanistic understanding of how increasing NAD availability via CD38 inhibition influences meibomian-gland tissue homeostasis.

Results

A vicious cycle between intracrine activity and inflammation in the meibomian gland

Tissue-intrinsic steroidogenesis (i.e., intracrine activity) via the enzyme 3 β -HSD is essential for maintaining normal meibomian gland tissue homeostasis^{10,31–33}. Our previous study showed that genetic deletion of *Hsd3b6*, the mouse enzyme for meibomian gland 3 β -HSD activity, reduced gene expression involved in cell proliferation in meibomian gland acinar cells, which mirrors the atrophy of the meibomian glands that occurred in *Hsd3b6*^{−/−} mice^{10,31}. However, the tissue-level alteration of the eyelid tissue of *Hsd3b6*^{−/−} mice remains to be examined. In order to address this question, whole eyelid tarsal plates— which contain both the meibomian glands and interstitial tissues surrounding the meibomian glands— were laser-microdissected and subjected to microarray (Fig. 1A, B). Volcano plot representations of *Hsd3b6*^{−/−} versus *Hsd3b6*^{+/+} expression data revealed significantly upregulated and downregulated transcripts ($|\log_2\text{FC}| > 0.32$, $P_{\text{adj}} < 0.1$) (see Fig. 1B). Gene Ontology (GO) analysis revealed that genes upregulated in *Hsd3b6*^{−/−} mice were mostly enriched in pathways related to immune responses, including *Irg1*, *Ccl3*, *Ccl9*, *Ccl2*, *Tnfrsf1b*, and *Tnf* (Fig. 1C,D), and down-regulated genes such as eosinophil-associated *Ear1* and *Ear2* also suggest altered immune homeostasis in *Hsd3b6*^{−/−} tarsal plates (Fig. 1B, blue plots). These data suggest that the depletion of meibomian gland local steroidogenesis results in inflammation in the tarsal plate.

Testosterone and 5 α -dihydrotestosterone (DHT) are the main active steroid hormones that the meibomian glands produce via 3 β -HSD-mediated intracrine activity^{10,31,32}. Given that androgens exert anti-inflammatory effects^{19,34–37}, these hormones may mediate the phenotype of the *Hsd3b6*^{−/−} tarsal plate. Supporting this idea, local supplementation of DHT into the eyelids of *Hsd3b6*^{−/−} mice normalized the level of increased mRNA expression of proinflammatory genes *Tnf* and *Ccl3* (Fig. 1E,D) compared to vehicle control ($P < 0.05$, paired t test, DHT-treated right eye vs. control left eye). Thus, eyelid steroid activity appears to play anti-inflammatory roles.

We next investigated the possibility of adverse effects caused by inflammation on meibomian gland activity. To do this, we induced inflammation in wildtype mouse eyelids (*Hsd3b6*^{+/+}, Fig. 1F). By local microinjection of lipopolysaccharide (LPS) in the eyelids, expected increased expression of inflammatory genes was observed (Fig. 1G: *Tnf*, *Il1b*, *Il6* and *Ccl3*). Associated with this alteration, meibomian gland intracrine activity monitored by 3 β -HSD activity was significantly reduced due to LPS treatment ($P < 0.05$, unpaired t test, vs. Veh treatment, Fig. 1H). These data thus point to a converse negative effect by inflammation on the meibomian tissue intracrine activity. We therefore inferred that a vicious cycle may underlie the phenotype of the meibomian gland atrophy, in which decreased meibomian intracrine activity leads to inflammation, and the resultant inflammation exacerbates the reduction in meibomian intracrine activity (Fig. 1I, schematic).

Inflammatory CD38–NAD axis as a mediator for intracrine–inflammation vicious cycle

Because meibomian gland intracrine activity depends on 3 β -HSD cofactor NAD availability and meibomian gland dysfunction is associated with decreased meibomian-gland NAD levels in mice¹⁰, we explored the possibility that NAD mediates the phenotype(s) that we observed in Fig. 1. Intracellular NAD levels are predominantly modulated by enzymes involved in NAD salvage biosynthesis, including nicotinamide (NAM) phosphoribosyltransferase (NAMPT) and nicotinamide mononucleotide (NMN) adenylyltransferase1-3 (NMNAT1-3), as well as enzymes involved in NAD consumption, including CD38, Sirtuins, and poly (ADP-ribose) polymerase (PARP)^{38–40} (see Fig. 2A). Among these enzymes, *Cd38* mRNA expression was most evidently altered by the absence of *Hsd3b6* (Fig. 2B, $P = 0.02$, *Hsd3b6*^{−/−} vs. *Hsd3b6*^{+/+}) with a fold-increase value of ~ 1.27 compared to *Hsd3b6*^{+/+} wildtype mice (see also Fig. 1B). *Parp2* and *Nampt* were mildly decreased in the absence of *Hsd3b6*, among others, while their fold-change (FC) values were

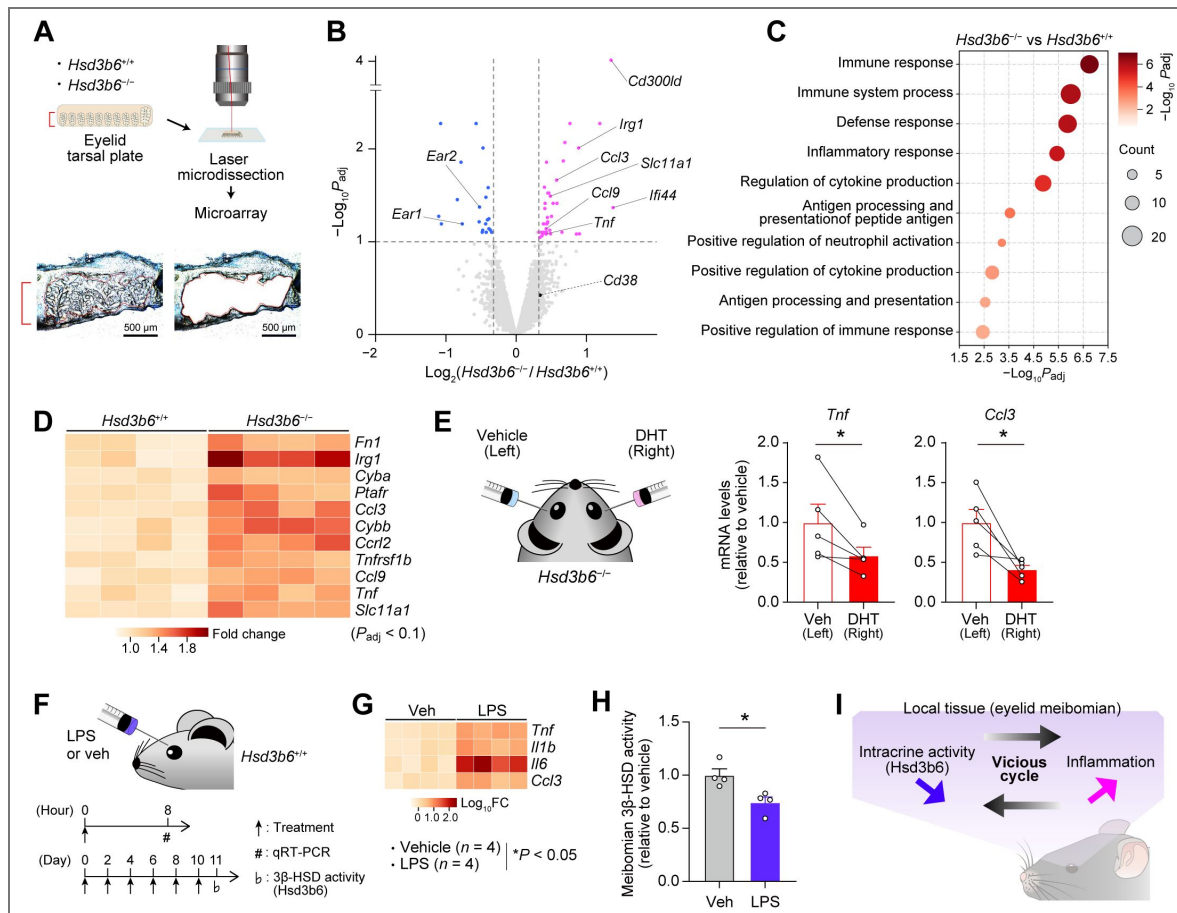


Figure 1. A vicious cycle between meibomian intracrine activity and inflammation.

(A) Laser-microdissection of eyelid tarsal plates of *Hsd3b6*^{+/+} and *Hsd3b6*^{-/-} mice. (B) Volcano plot of differentially expressed genes between *Hsd3b6*^{+/+} and *Hsd3b6*^{-/-} eyelid tarsal plates. $n = 4$ biologically independent samples for both genotypes. (C) Bubble plot showing GO terms for upregulated DEGs in *Hsd3b6*^{-/-} tarsal plates. (D) Heatmap showing genes categorized as inflammatory response in (C). $n = 4$ per genotype. (E) Effects of eyelid subcutaneous injection of DHT on *Tnf* and *Ccl3* expression in *Hsd3b6*^{-/-} tarsal plates ($n = 5$ mice). Left eyelids were treated with vehicle (Veh). (F) Experimental timeline for subcutaneous treatment of LPS for testing its effect on eyelid inflammatory cytokine gene expression and meibomian-gland intracrine activity. (G) Changes in expression of the inflammatory cytokine genes *Il1b*, *Il6*, *Tnf*, and *Ccl3* in (F). Values were determined by qRT-PCR ($n = 4$ for both Veh- and LPS-treated group). (H) Relative meibomian 3β-HSD activity in (F) ($n = 4$ per group). Values were determined by radioisotopic tracing of 3β-HSD activity using *in vitro* isolated tarsal plates. (I) Schematic illustration of decreased intracrine activity and inflammation, which form a vicious cycle in the eyelid. Data in (E) were analyzed using paired two-sided *t*-test; in (G), two-way analysis of variance (ANOVA) with Tukey's post hoc test; and in (H), unpaired two-sided *t*-test. Values represent means $\pm$ SEM; * $P < 0.05$.

~0.90 (Fig. 2B). Therefore, we sought to investigate CD38 further. Because CD38 is known to act as a key mediator for inflammation-associated NAD quantity modulation in the liver and adipose tissue^{41,42}, CD38 may be linked to the *Hsd3b6*^{-/-} inflammation phenotype in the eyelids. To explore the location of CD38 expression in the *Hsd3b6*^{-/-} tarsal plate, we performed immunofluorescence staining using anti-CD38 antibody (Fig. 2C). Coronal tarsal plate sections containing multiple separate meibomian-gland bunches revealed an accumulation of CD38-immunopositive cells in the interstitial tissue surrounding the meibomian glands in *Hsd3b6*^{-/-} mice (Fig. 2C,D). We also found that the macrophage marker CD11b stained these cells (Fig. S1), indicating an increased number of CD38-positive macrophages in the tarsal plate of *Hsd3b6*^{-/-} mice (Fig. 2D, vs. *Hsd3b6*^{+/+}). More importantly, accompanying this CD38 accumulation, meibomian gland NAD levels were decreased in *Hsd3b6*^{-/-} mice compared to normal *Hsd3b6*^{+/+} mice (Fig. 2E) and this reduction was ameliorated following administration of a specific CD38 inhibitor (78c) to the tarsal plates of *Hsd3b6*^{-/-} mice (see Fig. 2F). These data indicate that CD38-dependent decline of NAD occurs in the meibomian glands of *Hsd3b6*^{-/-} mice. The NAD values plotted were determined using laser-microdissected meibomian gland acinar cells (pmol mm⁻³ of tissue volume, 15.7 ± 1.0 for *Hsd3b6*^{+/+}; 11.8 ± 0.7 for *Hsd3b6*^{-/-}). Because administration of 78c, but not vehicle, restored meibomian NAD levels in *Hsd3b6*^{-/-} mice to levels comparable to those in *Hsd3b6*^{+/+} mice (14.8 ± 0.6 pmol mm⁻³ in 78c treatment; 10.8 ± 0.7 pmol mm⁻³ in Veh control, Fig. 2F), CD38 appears to be a major contributor to the NAD decline we observed, although minor contributions from other factors cannot be ruled out.

Following supplemental tissue local administration of DHT, the number of CD38⁺ cells in the eyelid tarsal plates was significantly decreased compared to vehicle control (Fig. 2G), and this was accompanied by increased meibomian NAD levels (Fig. 2H), supporting the notion of anti-inflammatory androgen action affecting CD38 accumulation and NAD decline in the tissue.

Moreover, following LPS treatment, which induced increased expression of *Cd38* in the tarsal plate (Fig. 2I)—while *Sirt1* and *Parp1* were decreased slightly and NAD biosynthesis genes (*Nampt* and *Nmnat1-3*) remained unchanged (Fig. 2I)—in normal *Hsd3b6*^{+/+} wildtype mice, we observed decreased meibomian gland NAD levels (*P* < 0.01, vs. Veh treatment, Fig. 2J), suggesting the participation of inflammatory CD38 in NAD level regulation. Consistent with this idea, 78c co-treatment reversed the reduction in NAD levels after LPS treatment (Fig. 2K). LPS-induced reduction in meibomian NAD-dependent 3β-HSD activity was also ameliorated by 78c treatment (Fig. 2L, *P* < 0.05, vs. Veh treatment; see also Fig. 1H). These data indicate that CD38 is a key mediator for inflammation-dependent modulation of local intracrine activity.

Aging involves CD38-dependent decline in meibomian gland intracrine activity

Since we so far obtained data from experimentally induced inflammatory conditions such as those due to genetic deletion of *Hsd3b6* and LPS administration, we next investigated whether physiological aging, characterized by subclinical chronic inflammation^{43,44}, affects CD38-modulated local intracrine activity (Fig. 3A).

We first asked whether age-associated CD38 accumulation occurs in human eyelid specimens obtained from postmortem donors (males, 41 years, 85 years; females, 34 years, 85 years). In the interstitial tissue surrounding the 3β-HSD-positive meibomian gland acini, CD38⁺ cells were observed in elderly donor samples (85-year-old male and female), whereas CD38⁺ cells were rarely observed in younger samples (41-year-old male and 34-year-old female) (Fig. 3A; Fig. S2), indicating a trend of CD38 accumulation in elderly subjects (Fig. 3B). Using mouse eyelid tissue specimens, we further examined whether CD38 accumulation occurs in mice due to aging. Immunofluorescence staining of eyelid tarsal plates of aged and young mice (2–6 vs. 20–24 months old, *n* = 7 for both age groups) revealed a significant increase in the number of CD38⁺ cells around the acini in aged mouse group (*P* < 0.001, Fig. 3C,D). Among the NAD salvage pathway genes examined (*Nampt*, *Nmnat1-3*, *Sirt1*, *Parp1*, and *Cd38*), only *Cd38* expression was considerably increased upon aging in the tarsal plate, as assessed by qRT-PCR (Fig. 3E). Western blotting confirmed CD38 protein accumulation in the tarsal plate of aged (20 months) mice, with

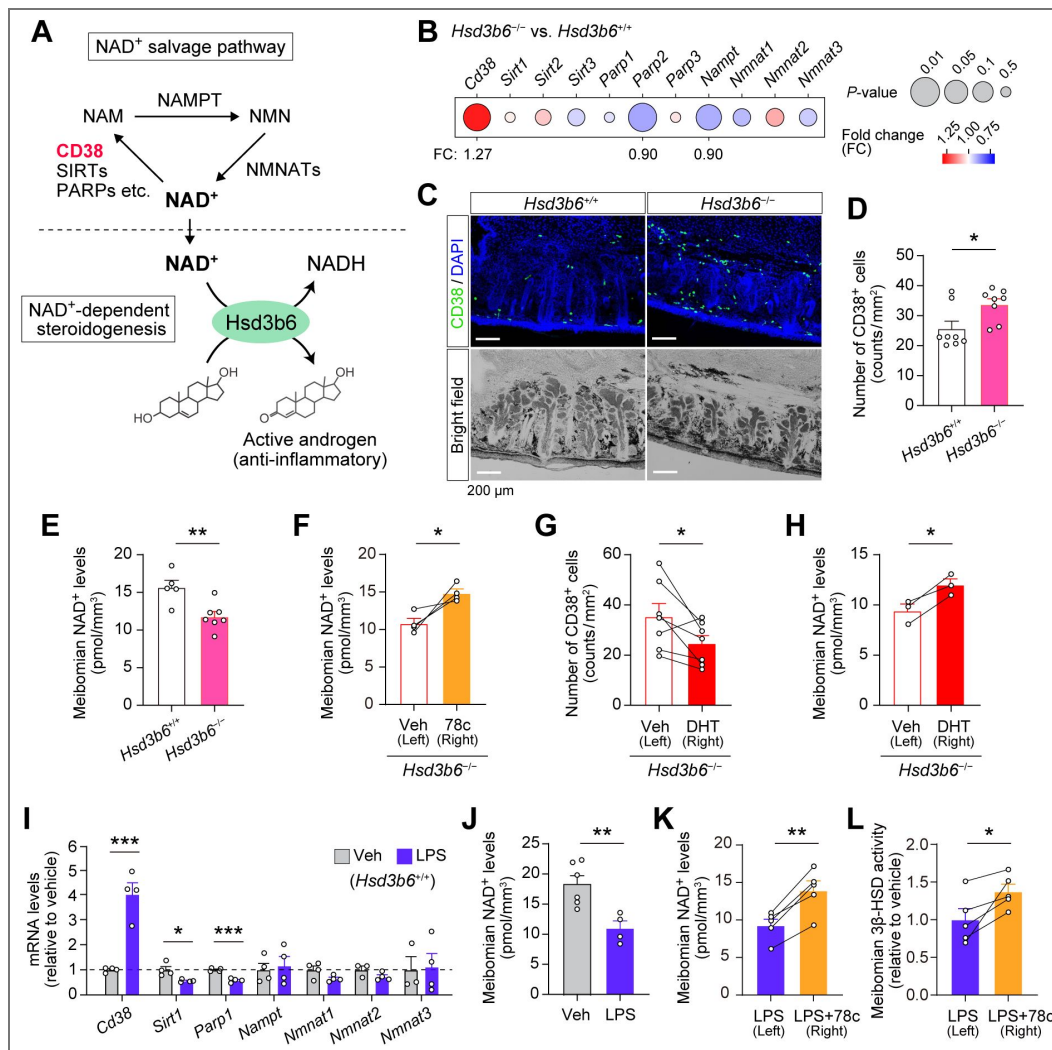


Figure 2. CD38 mediates NAD depletion-driven vicious cycle in the meibomian gland.

(A) Schematic of the NAD salvage pathway and NAD-dependent enzymatic activity of Hsd3b6. (B) *Hsd3b6*^{-/-} versus *Hsd3b6*^{+/+} expression of genes involved in the NAD salvage pathway. The color of circles indicates fold change and the size indicates P-value. Data are based on Fig.1B. (C) Representative immunofluorescence images of CD38 in *Hsd3b6*^{+/+} and *Hsd3b6*^{-/-} eyelid coronal sections. The sections were visualized by DAPI staining and bright field microscopy. (D) The number of CD38 positive cells around meibomian gland acini in *Hsd3b6*^{+/+} and *Hsd3b6*^{-/-} mice. *n* = 8 eyelids per genotype. (E) Meibomian gland NAD levels in *Hsd3b6*^{+/+} (*n* = 5) and *Hsd3b6*^{-/-} (*n* = 7) mice. (F) Meibomian gland NAD levels in *Hsd3b6*^{-/-} mice after eyelid subcutaneous injection of 78c (*n* = 4 mice). Left eyelids were treated with Veh. (G) The number of CD38 positive cells around meibomian gland acini in *Hsd3b6*^{-/-} mice after eyelid subcutaneous injection of DHT (*n* = 7 mice). Left eyelids were treated with Veh. (H) Meibomian gland NAD levels in *Hsd3b6*^{-/-} mice after eyelid subcutaneous injection of DHT (*n* = 3 mice). Left eyelids were treated with Veh. (I) Relative mRNA levels of *Cd38*, *Sirt1*, *Parp1*, *Nampt*, *Nmnat1*, *Nmnat2*, and *Nmnat3* in the tarsal plate of wildtype mice after eyelid injection of Veh (*n* = 3-4) or LPS (*n* = 4). (J) Meibomian NAD⁺ levels in wildtype mice after treatment of Veh (*n* = 6) or LPS (*n* = 4). (K) Effects of 78c on meibomian NAD⁺ levels in wildtype mice treated with LPS (*n* = 5). Left eyelids received Veh without 78c after eyelid injection of LPS. (L) Effects of 78c on meibomian 3β-HSD activity in wildtype mice treated with LPS (*n* = 5). The eyelid tarsal plates from LPS-treated mice were incubated in culture with or without 78c. Data in (D), (E), (I) and (J) were analyzed using unpaired two-sided *t*-test; in (F), (G), (H), (K) and (L), paired two-sided *t*-test. Values are means ± SEM; **P* < 0.05, ***P* < 0.01, ****P* < 0.001.

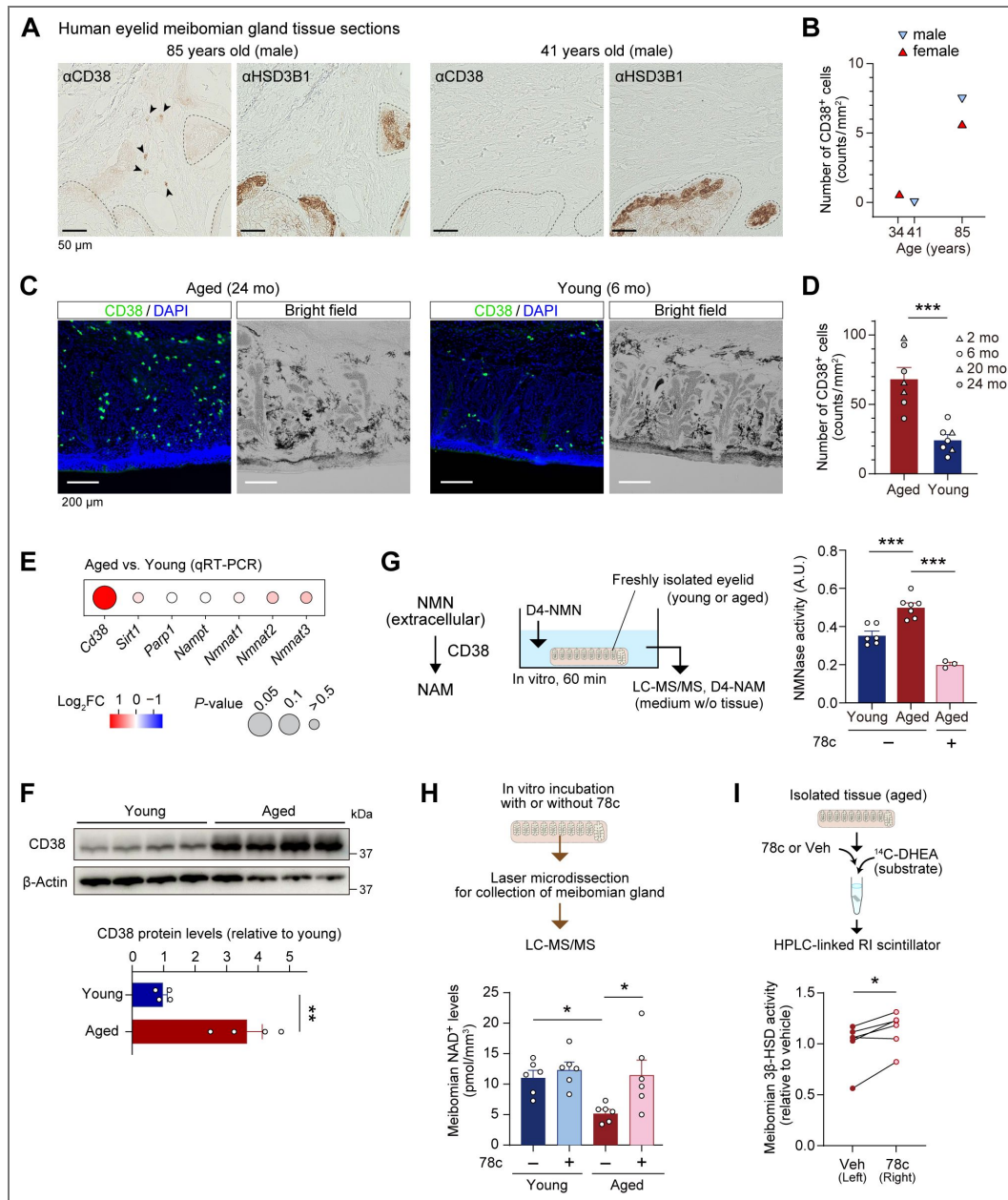


Figure 3. Increased CD38-driven eyelid intracrine-activity decline in aged mice.

(A) Human eyelid tissue sections stained for CD38. Shown are a pair of serial sections from 85-year-old and 41-year-old male eyelid specimens, stained for CD38 and HSD3B1. Dashed lines delineate HSD3B1-positive meibomian acini. CD38-immunopositive cells are pointed by arrowheads. (B) The number of CD38 positive cells around meibomian acini in human eyelids (41-year-old female, 85-year-old male and 85-year-old female). (C) Representative immunofluorescence images of CD38 in aged (24 mo) and young (6 mo) eyelid sections. The sections were visualized by DAPI staining and bright field microscopy. (D) The number of CD38 positive cells around meibomian acini in young (2-6 mo) and aged (20-24 mo) mice ($n = 7$ for each group). (E) Age-associated gene expression changes in the NAD-salvage pathway. Tarsal plates of aged and young mice were subjected to gene-specific qRT-PCR. $n = 6-12$ per group. (F) Immunoblotting of CD38 in young (2 mo) and aged (20 mo) eyelid tarsal plates ($n = 4$ for each group). β -Actin was used as a loading control. (G) Quantification of ecto-enzymatic NMNase activity in isolated tarsal plates. The rate of conversion from D4-NMN to D4-NAM was quantified for young (2-6 mo, $n = 6$), aged (21 mo, $n = 7$), and 78c-treated aged (21 mo, $n = 3$) mouse tarsal plates. (H) Meibomian gland NAD levels in young (2-5 mo) and aged (20 mo) mouse eyelids after in vitro incubation with 78c or Veh ($n = 6$ for each group). (I) β 3-HSD activity in the eyelid tarsal plates incubated in culture with or without 78c ($n = 6$ eyelids per condition, 24-25 mo). Statistics in (D), (E), and (F), unpaired two-sided t -test; in (G), one-way ANOVA followed by Bonferroni's test; in (H), two-way ANOVA followed by Tukey's test; and in (I), paired two-sided t -test. Values are means $\pm$ SEM; * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. mo, months old.

an approximately 3.7-fold increase compared to young (2 months) mice (Fig. 3F). A different monoclonal antibody for the mouse CD38 also reproduced the increase of CD38 immunopositive cells in the tarsal plate of aged mice (Fig. S3, immunofluorescence staining).

Having observed age-associated increase in *Cd38* mRNA/protein expression, we next asked whether this has a functional influence(s) on meibomian glands. We previously showed that the levels of NAD in the meibomian gland cells decrease with age¹⁰. However, the underlying mechanism has been unclear. Because CD38⁺ inflammatory cells degrade extracellular NMN through their ecto-enzymatic activity and thereby reduce tissue NMN-NAD availability in the liver, muscle, and adipose tissue^{41,45,46}, we assumed a similar CD38-dependent mechanism in the tarsal plate. Aligned with this hypothesis, freshly isolated eyelid tarsal plates from young and aged mice, incubated *in vitro* in Hanks' Balanced Salt solution containing deuterium (D4)-labeled NMN, revealed NMNase activity (conversion from D4-NMN to D4-NAM) that was increased by aging and suppressed by 78c treatment (see Fig. 3G, $P < 0.001$, vs. young, vs. 78c). Moreover, we measured the total NAD levels in the meibomian gland cells. To do this, we isolated the meibomian gland cells by laser-microdissection after *in vitro* incubation of freshly isolated mouse tarsal plates in the presence or absence of 78c (Fig. 3H). The levels of NAD in aged mouse samples were decreased compared with those in young control samples in the absence of 78c (Fig. 3H, $P < 0.05$). Notably, meibomian gland NAD levels in aged mouse samples were restored to the levels that were comparable to those in young mouse samples after 78c treatment (Fig. 3H, $P < 0.05$), indicating a critical involvement of CD38 activity in the downregulation of meibomian gland NAD levels by aging. On the other hand, CD38 inhibition (by 78c) had no profound effect on the meibomian NAD levels in young mouse samples, consistent with the minimal *Cd38* mRNA/protein expression in the tarsal plate of young mice (Fig. 3H and 3C-F). Alongside the restoration of meibomian NAD availability, 3 β -HSD (which is selectively expressed in the meibomian gland cells) activity in the tarsal plates was increased following 78c treatment in aged mouse tissue samples under the same *in vitro* conditions (see Fig. 3I, $P < 0.05$, paired *t* test). Taken together, these data indicate that physiological aging involves CD38-dependent downregulation of NAD availability and intracrine activity in the meibomian glands.

In vivo local treatment of the eyelid with 78c

To test the potential beneficial effect of 78c on the meibomian glands, we next examined the effect of eyelid local treatment of 78c in aged mice (Fig. 4). We first assessed the meibomian gland NAD levels. We used aged mice that received 78c to the eyelid of the right eye (micro-injections of 78c were performed at 4 h and 20 h before tissue sampling). As for the left eye of the mice, we treated vehicle (Fig. 4A, a cartoon). A statistically significant increase in the meibomian gland NAD level was observed in the right eye (78c treatment) compared to the left eye (paired *t* test, $P < 0.05$, Fig. 4B). Moreover, to test the effect of 78c on the meibomian gland morphology, we next performed a longer-term drug treatment where 78c was injected once every 3 days for 7 weeks. We observed that the eyelids treated with 78c had significantly enlarged meibomian glands, as revealed by whole-mount eyelid meibomian lipid staining (see Fig. 4C, $P = 0.04$, vs. left eyes treated with vehicle). These data extend the evidence for the beneficial effect of 78c from *in vitro* treatment (see *ex vivo* culture; Fig. 3) to *in vivo* treatment conditions, albeit with only borderline statistical significance observed in the *in vivo* setting.

As a logical possible improvement, we considered that treatment of NMN in addition to 78c might have a more profound beneficial effect because 78c would also reduce degradation of exogenously supplied NMN. To explore this possibility, we first quantified meibomian NAD levels following an additional treatment of NMN after the administration of 78c (mice additionally received an eye-drop instillation of NMN at 1 h before tissue collection). Notably, the 78c-NMN cotreatment resulted in a greater increase in meibomian NAD levels compared to either 78c or NMN treatment alone (pmol mm⁻³ of tissue volume, 17.5 $\pm$ 1.0 for 78c; 18.5 $\pm$ 1.0 for NMN; 22.3 $\pm$ 0.8 for 78c+NMN, Fig. 4B). Moreover, after a longer-term cotreatment with 78c and NMN (in addition to receiving 78c treatment, mice received NMN instillation 3 times every 2 days), we observed a significant

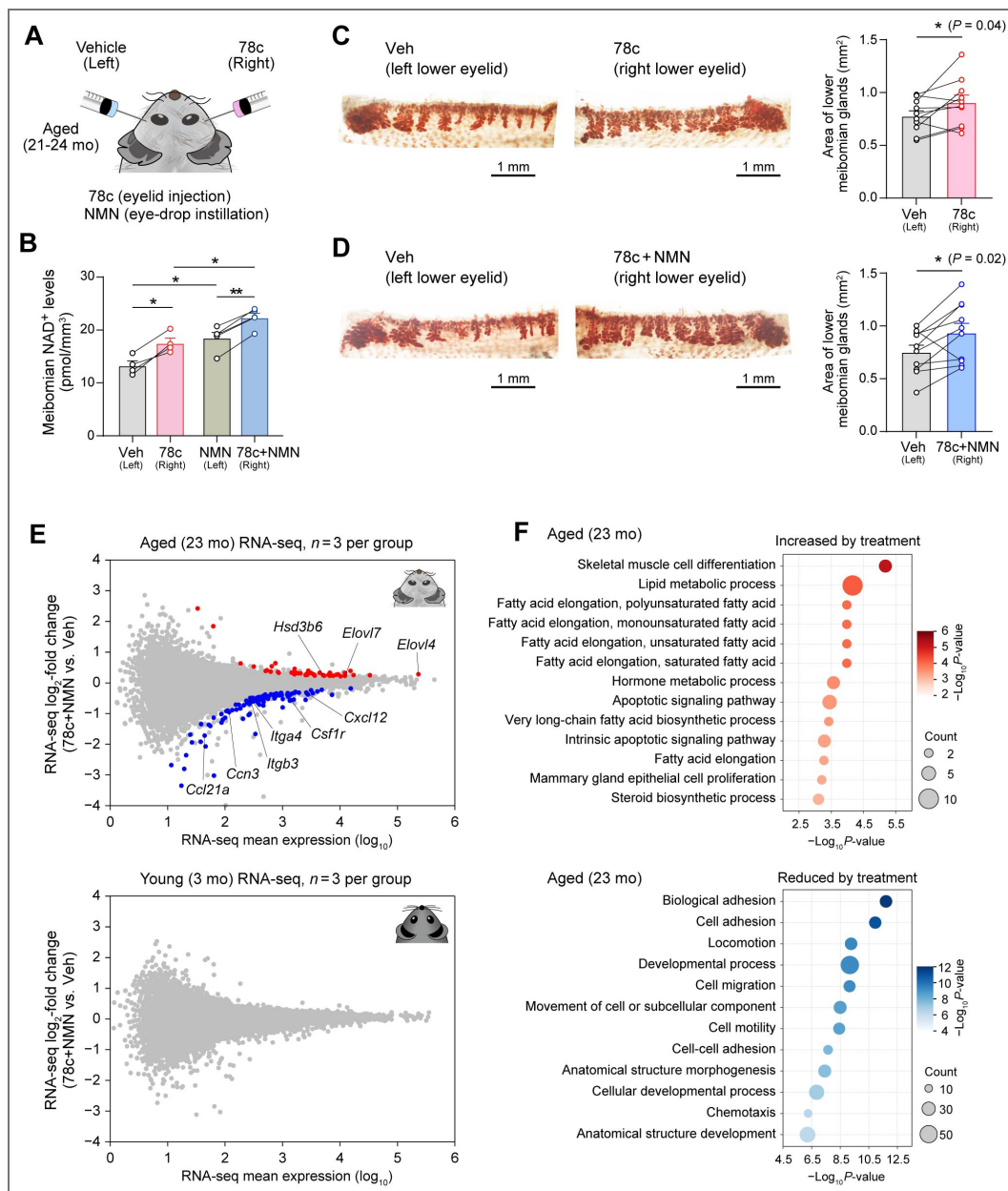


Figure 4. Age-associated meibomian gland phenotype after 78c-NMN in vivo treatment.

(A) Schematic of 78c and NMN treatment. 78c was subcutaneously injected into lower eyelid and NMN was administered via eye-drop. Left eye served as a treatment control. (B) Meibomian gland NAD levels in aged (21 mo) mice treated with Veh (*n* = 4), 78c (*n* = 4), 78c plus NMN (*n* = 5), or NMN (*n* = 5). Mice received two doses of Veh or 78c 20 h and 4 h prior to euthanasia. One dose of NMN was given by instillation at 1 h before euthanasia. (C-D) Meibomian gland morphology in the lower eyelids of aged (24 mo) mice treated with 78c (C) or 78c plus NMN (D). Left eyes were treated with Veh. The graph shows the size of the meibomian gland area (*n* = 10 mice per condition). Treatment was performed for 7 w, with administration of 78c once every 3 d and NMN three times every 2 d. (E) RNA-seq data showing 78c+NMN-versus Veh-treated eyelid tarsal plate transcriptome, presented in MA-plot format (*n* = 3 biologically independent samples for each group; each sample was generated using a pool of 3 lower eyelids) for aged and young mice. Plots in red and blue represent transcripts significantly upregulated and downregulated by 78c+NMN treatment, respectively ($P_{\text{adj}} < 0.1$). Treatment was performed for 4 w. (F) Bubble plot showing GO terms for upregulated (top) and downregulated (bottom) DEGs in 78c+NMN treated tarsal plates of aged mice. Data in (B) were analyzed using paired or unpaired two-sided *t*-tests with FDR correction; in (C) and (D), paired two-sided *t*-test. Values are means $\pm$ SEM; **P* < 0.05, ***P* < 0.01.

enlargement of the meibomian gland size with higher statistical confidence compared to 78c treatment alone (Fig. 4D [↗](#), $P = 0.02$, vs. left eyes treated with vehicle; see also Fig. 4C [↗](#)), suggesting an improved beneficial effect of 78c-NMN cotreatment on meibomian gland atrophy.

To understand the effect of 78c-NMN treatment from the perspective of gene expression in the meibomian gland, we performed comparative RNA-seq analysis using young (3 months) and aged (23 months) mice that were treated with 78c-NMN or vehicle administration for 4 weeks (the right eyes received 78c-NMN and the left eyes received vehicle; $n = 9$ mice for both age groups; each RNA sample was pooled from 3 independent mice, resulting in $n = 3$ RNA samples for RNA-seq). From these data (see Fig. 4E [↗](#)), we observed that, in aged mice, the eyelid gene transcriptome was altered due to 78c-NMN treatment, with 42 differentially expressed genes (DEGs) upregulated and 99 genes downregulated (Fig. 4E [↗](#), red, upregulated; blue, downregulated). GO analysis revealed that upregulated genes were enriched in pathways related to lipid metabolic processes, including *Elovl4*, *Elovl7*, and *Hsd3b6*, while down-regulated genes were associated with cell adhesion, migration, and chemotaxis, including *Ccn3*, *Itgb3*, *Itga4*, *Cxcl12*, *Ccl21a* and *Csf1r* (Fig. 4E, F [↗](#)). Notably, the elevation of *Elovl4* and *Elovl7*—elongases responsible for the synthesis of very long-chain fatty acids that play a key role in meibomian tissue homeostasis^{47,48}—as well as *Hsd3b6* (3 β -HSD and a marker for the meibomian acinar cells) aligned with an improved size of the meibomian gland. In contrast, the same treatment in young mice did not significantly alter gene expression profiles (see Fig. 4E [↗](#), young), a result consistent with that observed in previous studies^{46,49,50} showing minimal impact of 78c (or NMN) on young mice compared to aged mice in terms of multiple physiological parameters including glucose homeostasis, exercise tolerance, and cardiac function.

A potential virtuous cycle in NAD boosting in the eyelid meibomian gland

Finally, returning to our initially proposed intracrine-inflammation “vicious” cycle model, we wondered whether the phenotype induced by the NAD boosting may involve the reversal of this cycle. We considered that in association with the restoration of meibomian gland NAD availability, age-associated increment of CD38⁺ cell accumulation would be suppressed due to anti-inflammatory action of intracrine activity. Indeed, the number of CD38⁺ cells around the meibomian gland acini was significantly decreased following 4 weeks of 78c-NMN treatment compared to vehicle treatment (Fig. 5A [↗](#), $P < 0.05$), which is consistent with the reduced expression of *Cd38* in RNA-seq data (Fig. 5A [↗](#), *Cd38* mRNA, vs. Veh). Moreover, in the absence of intact intracrine activity, the number of CD38⁺ cells was not reduced following the same 78c-NMN treatment; thus, intracrine activity is essential for eliciting the reduction in CD38⁺ cells (Fig. 5B [↗](#), *Hsd3b6*^{−/−}). Moreover, by using transcriptome data, we searched for a shared gene expression signature between the effect of intracrine activity deficiency (*Hsd3b6*^{−/−}) and the effect of 78c-NMN treatment (Fig. 5C [↗](#)). This led to the identification of 94 overlapped genes displaying up-regulation by *Hsd3b6* deletion and downregulation following 78c-NMN treatment, with the results of GO analysis revealing a pronounced enrichment for immune-related pathway (Fig. 5C [↗](#)). These transcriptomic changes, in addition to the reversal of CD38⁺ cell accumulation, suggest the involvement of anti-inflammatory action of intracrine activity (*Hsd3b6*) in the effects of NAD augmentation on meibomian glands.

Discussion

Vicious cycles amplify biological malfunction, forming the underlying mechanism of various diseases^{1–6}. In this context, in our study, we provide evidence to support the presence of vicious cycle for the pathology of meibomian gland dysfunction, a chronic eyelid disorder known as the most prevalent form of evaporative dry eye disease^{51,52} with currently no established (or approved) pharmacological therapy^{53–56}. For the key component of our vicious cycle model, we identified a previously unknown mutual interaction between meibomian gland intracrine activity and inflammation in the tarsal plate. More specifically, we found that the loss of intracrine activity leads to inflammatory upregulation in the tarsal plate. Conversely, induced inflammation

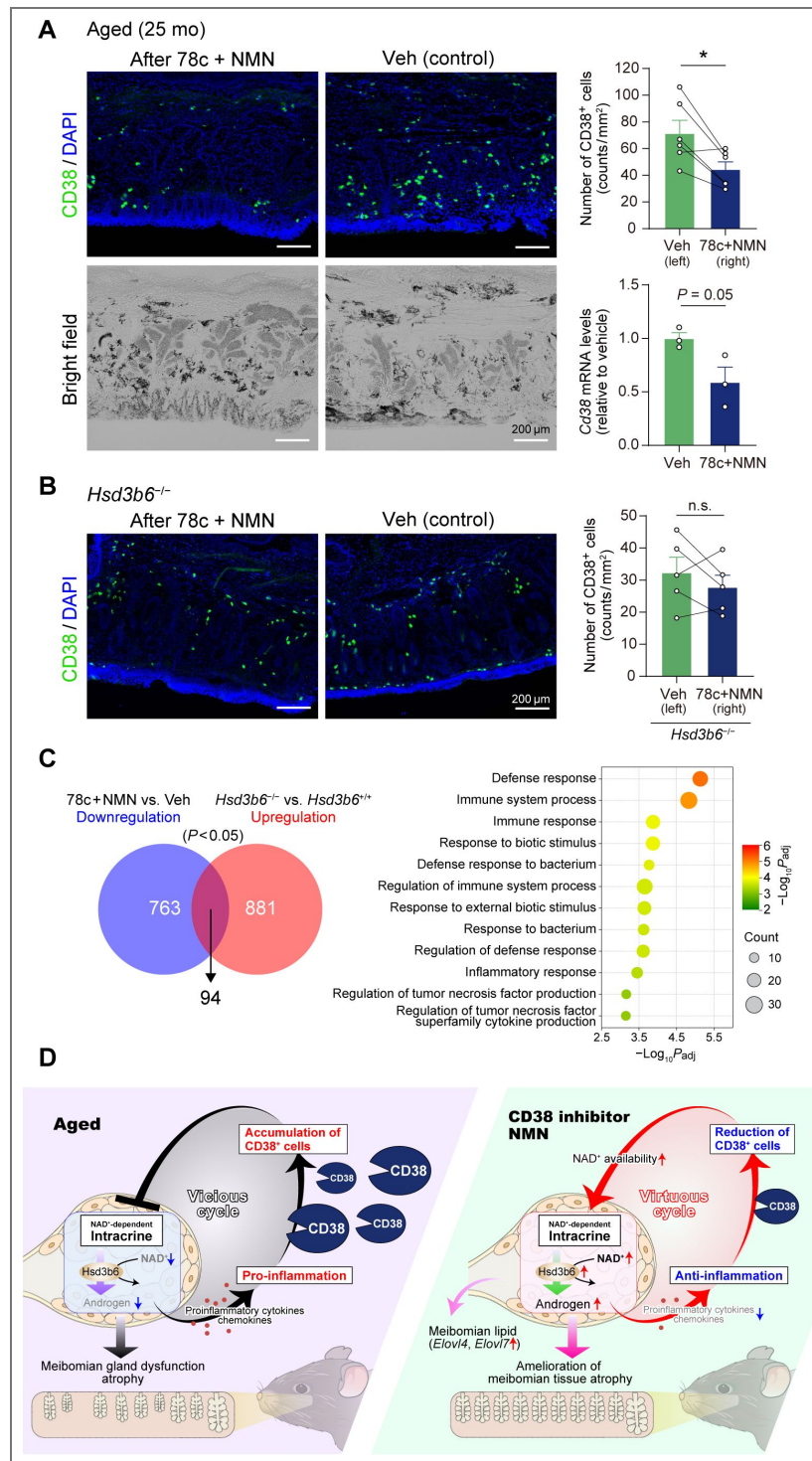


Figure 5. Reduced inflammation due to 78c+NMN treatment, a potential virtuous cycle.

(A) Representative immunofluorescence images of CD38 in Veh-treated or 78c+NMN treated eyelids of aged mice. The upper graph shows the number of CD38 positive cells around meibomian acini in each condition (*n* = 6 mice, 25 mo). The lower graph indicates RNA-seq data for *Cd38* mRNA in the tarsal plate in Fig. 4E. Treatment was performed for 4 w. (B) Representative immunofluorescence images of CD38 in Veh-treated or 78c+NMN treated eyelids of *Hsd3b6*^{-/-} mice. The graph shows the number of CD38 positive cells around meibomian acini in each condition (*n* = 5 mice, 5-10 mo). Treatment was performed for 4 w. (C) Venn diagram showing the overlap between downregulated genes by 78c+NMN treatment and upregulated genes upon *Hsd3b6* deletion. The shared DEGs were subjected to GO term enrichment analysis. (D) A schematic model of local circular circuits in the eyelids. A conversion from a pathologic vicious cycle into a therapeutic virtuous cycle may underlie 78c+NMN's effects on MGD. Data in (A, upper) and (B) were analyzed using paired two-sided *t*-test; in (A, lower), unpaired two-sided *t*-test. Values are means ± SEM; **P* < 0.05.

suppresses NAD-dependent 3β -HSD activity through CD38⁺ cell accumulation, thereby further diminishing the normal anti-inflammatory intracrine activity of the meibomian glands. Our data also show that NAD augmentation by 78c treatment, in addition to NMN, can liberate the meibomian gland from this pathological cycle. We found that, in association with a restoration of the meibomian gland tissue size, the number of CD38⁺ cells surrounding the meibomian glands was reduced following administration of 78c and NMN to the eyelids. Thus, rather than exerting a simple NAD boosting effect, 78c-NMN treatment appears capable of breaking the pathological vicious cycle to enter a reverse positive loop for enhancing the recovery of meibomian gland tissue homeostasis. Our data and model (Fig. 5D [↗](#)) thus may contribute to a mechanism-based understanding of the pathology of meibomian gland dysfunction and its potential reversibility.

Intracrine activity refers to the activity of locally synthesized hormones that exert effects in the same cells in which they are produced^{7–9}, in contrast to the endocrine hormones that act on distal target tissues. In the meibomian glands, the acinar cells themselves abundantly express androgen receptors (ARs)^{57,58}. In this unique situation, we have identified an anti-inflammatory contribution of meibomian gland intracrine activity in the eyelid, leading to the revision of the roles of intracrine activity. Combined with our previous finding¹⁰ that demonstrated the direct positive effect of intracrine activity on cell proliferation of meibomian acinar cells, our data expand the relevance of intracrine activity to both direct and indirect (secondary) support to the acinar cells through maintaining NAD availability via anti-inflammatory suppression of CD38⁺ cell accumulation in the eyelid tarsal plate. These data contribute to understanding of integrative roles of meibomian intracrine activity. We should describe that the molecular basis of anti-inflammatory action of intracrine activity remains unclear from our study. According to the theory of intracrine biology, the primary site of action would include the meibomian gland acinar cells *per se*. Previous studies using immortalized human meibomian gland epithelial cells (HMGECS)⁵⁹ support this notion. Transcriptome studies with and without androgen treatment indicate that androgen has a role in reducing expression of numerous immune-related genes, such as those associated with antigen processing and presentation, innate and adaptive immune responses, chemotaxis, and cytokine production, in HMGECS³⁷. These data suggest that androgen exerts its anti-inflammatory effect at least partly via acting directly on the meibomian acinar cells. With that said, other mechanisms are also possible. For example, ARs are likely expressed in macrophages^{60,61}, and there are several studies showing the alteration of macrophages' immune responsiveness by androgen treatment^{62,63}, although the exact molecular pathway by which androgen alters macrophages' immune responsiveness has not yet been fully elucidated⁶⁴. It is conceivable that multiple unknown parallel pathways involving meibomian gland cells and other cells require investigation to understand the anti-inflammatory actions of intracrine activity.

The enzymatic activity of 3β -HSD depends on the availability of NAD. The values of the Michaelis constant (K_M) for NAD of the mouse and human meibomian gland 3β -HSD (i.e., Hsd3b6 for mouse; HSD3B1 for human) were 20 μ M and 34 μ M, respectively^{10,30}, and these values fall in a physiological range of cellular NAD concentration. Of note, 78c treatment *in vitro* (Fig. 3H [↗](#)) caused meibomian gland NAD levels in aged mice to increase from 5.3 ± 0.6 to 11.6 ± 2.4 μ M (determined using tissue volume). *In vivo* eyelid local treatment of 78c/NMN also increased meibomian NAD levels from 13.3 ± 0.9 to 22.3 ± 0.8 μ M in aged mice (Fig. 4B [↗](#), means $\pm$ sem). These data reinforce the notion that NAD is a rate-limiting coenzyme for the 3β -HSD to exert its role. Our previous study using cultured human steroidogenic cells (H295R)¹⁰ demonstrated that pharmacological inhibition and upregulation of NAD biosynthesis in these cells led to corresponding inhibition and elevation of intracellular 3β -HSD activity, supporting the importance of NAD bioavailability as a determinant of 3β -HSD activity. In this context, in our current study, we have identified a contribution of the NAD/NMN-degrading enzyme CD38 to the reduction of NAD bioavailability in the meibomian gland. Our data thus expanded the research field of CD38-mediated pathology to include the meibomian gland dysfunction that involves NAD-dependent enzyme for local steroid biosynthesis. It is worth noting that the critical involvement of CD38 in NAD availability has already been documented for several age-associated diseases, including obesity-related disorders^{46,65,66}, neurodegenerative disorders^{67–69}, cardiovascular dysfunction^{70–72}, gonadal dysfunction^{73,74}, and hematopoietic stem cell deterioration⁷⁵, each involving distinct

NAD-dependent molecular pathways. In the case of meibomian pathology, we focused on the NAD dependency of 3 β -HSD — however, given the pleiotropic roles previously determined for NAD, other mechanisms may also contribute. For example, 3 β -HSD (Hsd3b6 for mouse; HSD3B1 for human) may be directly regulated by sirtuin-mediated deacetylation—an NAD-dependent process associated with aging⁴⁰—which warrants further investigation. Moreover, considering that NAD contributes to circadian clock reprogramming⁷⁶, improved circadian clock function may secondarily enhance the activity of 3 β -HSD. It is also likely that NAD augmentation in the eyelids and eyes influences cell types other than meibomian gland cells, including cells in the conjunctiva, cornea, lacrimal gland, and other related cells^{77,78}. Understanding how these mechanisms intersect with the intracrine–inflammation vicious cycle in the meibomian glands will be an important direction of our future study.

In this study, we reported a previously uncharacterized pharmacological potential of CD38 inhibition by 78c in the eyelid. Nevertheless, several limitations need to be considered from a clinical pharmaceuticals perspective. Most notably, the present study relied on direct local injections of 78c into the eyelid, and our attempt at drug delivery through simple eye-drop instillation of 78c did not sufficiently elevate meibomian NAD levels. This indicates that, for therapeutic translation, formulation strategies that ensure efficient penetration of CD38 inhibitors—or NAD precursors—into the tarsal plate will be essential. Overcoming this critical barrier will be required before 78c or related agents can be developed as practical ophthalmic medicines. Despite these limitations, the ability of CD38 inhibition to restore intracrine activity and suppress inflammatory CD38⁺ cell accumulation suggests that breaking the pathological loop may, in turn, initiate a virtuous cycle that helps sustain meibomian gland homeostasis. These pharmaceutically actionable insights merit further investigation toward the development of mechanism-guided therapy for meibomian gland dysfunction.

Materials and methods

Animals

Young and aged C57BL/6J male mice were purchased from CLEA Japan Inc. *Hsd3b6*-null (*Hsd3b6*^{−/−}) mice were bred and genotyped as described previously¹⁰. Mice were maintained under a regular 12-h light/12-h dark cycle (lights on 8:00 and off at 20:00) with free access to food and water as described⁷⁹. All animal experiments were conducted in compliance with the Ethical Regulations of Kyoto University and performed under protocols approved by the Animal Care.

Human eyelid specimens

Formalin-fixed, paraffin-embedded human eyelid samples were obtained from postmortem donors, including 41-year-old male, 34-year-old female, 85-year-old male, and 85-year-old female, who had no indications of symptoms for meibomian gland dysfunction (MGD) or dry eye disease (DE). All samples were purchased from Science Care (Phoenix, Arizona, USA).

Pharmacological treatments of eyelids

DHT (Tokyo Chemical Industry) was resolved in 5% DMSO–95% saline and subcutaneously injected into eyelids of *Hsd3b6*^{−/−} mice (15 ng/eyelid) once daily for six consecutive days.

LPS (Sigma) was reconstituted in saline and injected subcutaneously (5 μ g/eyelid) 8 h before tissue collection for qRT-PCR analysis. LPS was administered every other day for a total of three or six doses before the measurement of meibomian NAD⁺ concentration or meibomian 3 β -HSD enzymatic activity, respectively. 78c (BLDpharm) was reconstituted in 5% DMSO, 15% PEG400, and 80% of 15% hydroxypropyl- γ -cyclodextrin (in citrate buffer pH 6.0) and injected into lower eyelids (10 μ g/eyelid). For NMN treatment, PBS containing 5% NMN was administered via instillation as described¹⁰ (5 μ l/eye). For the short-term treatment of 78c (and NMN) for NAD⁺ measurement, mice received two doses of 78c or vehicle (at 20 h and 4 h before tissue collection) with or without

one dose of NMN (1 h before tissue collection). For the long-term treatment of 78c and NMN, mice received 78c once every 3 days and NMN three times every 2 days for a total of 4 weeks (for RNA-seq and CD38 accumulation test) or 7 weeks (for whole-mount meibomian gland morphology test).

Tissue sampling

Whole eyelid tarsal plates were used for 3β -HSD enzymatic activity measurement, ectoenzymatic NMNase activity measurement, immunoblotting, and specified qRT-PCR. The surgery of the tarsal plate was conducted as described¹⁰. In brief, this procedure involved making a small incision near the inner corner of the eyelid, separating skin/subcutaneous tissue from the inner to outer aspect of the lid, and then removing skin from the tarsal plate by cutting at the mucocutaneous junction. The isolated tarsal plates were collected into either medium or sampling buffer depending on the experiments. In experiments needing collection of meibomian-gland cells, including those for meibomian NAD measurement, microarray, RNA-sequencing and specified qRT-PCR, laser-microdissection was employed as previously described^{10,80}. In brief, cryosections (20 μ m thick) of fresh-frozen eyelids were mounted on a POL-membrane slide (Leica). For RNA analysis, sections were fixed in ice-cold ethanol-acetic acid mixture (19:1) for 2 min and stained with 0.05% toluidine blue. Meibomian glands were excised by laser microdissection and collected into Trizol reagent (Invitrogen) using a LMD7000 system (Leica). For the measurement of NAD concentration, sections were fixed in ice-cold 50% ethanol for 30 s and meibomian glands were micro-dissected into 50% methanol. The total size of the areas of the meibomian glands that were captured by laser microdissection was quantified for each sample using the Leica LMD software (Leica) for data normalization.

Radioisotopic measurement of 3β -HSD activity

The quantification of 3β -HSD enzymatic activity was described previously^{10,32}. Briefly, tarsal plates were freshly isolated from mice; then, the tissues were cut into slices (~0.3 mm thick) with a tissue chopper (McIlwain Lab) and immediately transferred into preaerated (5% CO₂, 95% O₂) Hank's balanced salt solution (HBSS) containing [4-¹⁴C]-dehydroepiandrosterone (3.6 μ M, American Radiolabeled Chemicals, Inc), dutasteride (10 μ M, Cayman), fadrozole hydrochloride (10 μ M, Sigma), and 4% propylene glycol. After 90 min at 37°C, the assay medium was collected into ethyl acetate (1:5). Where indicated, tissue slices were pretreated with 78c (10 μ M, Sigma) in HBSS for 30 min before assay. The extracts were evaporated to dryness under a nitrogen stream at 75°C. The dry extracts were reconstituted into 40% acetonitrile and subjected to HPLC-connected RI scintillation as described³².

LC-MS/MS for meibomian NAD quantification

LC-MS/MS was performed using a Shimadzu Nexera UHPLC/HPLC System (Shimadzu) coupled to a LC-MS-8030 triple quadrupole mass spectrometer (Shimadzu) as described^{10,32}. The meibomian gland cells isolated into 50% methanol were mixed with chloroform (1:1) followed by sonication. After centrifugation, supernatant aqueous samples were filtered (0.22 μ m) and freeze-dried. The samples were resolved in water and separated through a COSMOSIL Packed Column 5C18-PAQ (2.0 mm $\times$ 150 mm). NAD was detected using the multiple reaction monitoring mode and identified by comparison of its LC retention time and MS² fragmentation pattern (m/z 663.85 $>$ 136.05) with those of the authentic standard. NAD levels were quantified based on the peak area compared to a standard curve. The values of NAD were normalized to the tissue volume (mm³) captured by laser microdissection.

Ecto-enzymatic NMNase activity measurement

Freshly isolated tarsal plates were incubated in vitro in preaerated HBSS containing 100 μ M D4-NMN (Toronto Research Chemicals) in the presence or absence of 78c (Sigma, 10 μ M) with a mixture of NMNAT inhibitor gallotannin (Sigma, 100 μ M), CD73 inhibitor APCP (Jena Bioscience, 500 μ M) and NNMT inhibitor NNMTi (Sigma, 100 μ M) at 37°C for 1 h. Following incubation, the medium (w/o tissue) was mixed with methanol and chloroform (1: 1:2). The samples extracted

were then filtered (0.22 μ m), lyophilized, and analyzed by LC-MS/MS. Quantification was performed through multiple reaction monitoring mode at m/z transitions of 339.10 > 127.10 for D4-NMN and 127.00 > 84.05 for D4-NAM.

Whole-mount meibomian gland visualization

Whole-mount lipid staining of meibomian glands was performed as described previously^{10,31}. Briefly, the whole skin around the eye was excised and fixed with 4% paraformaldehyde and 1% glutaraldehyde in PBS for 24 h at 4°C; then, the specimens were bleached in 3% hydrogen peroxide for 24 h at 37°C and post-fixed with the same fixative, followed by dehydration in 50% and 70% ethanol and staining with Sudan IV (Nacalai)-saturated 70% ethanol containing 2% NaOH. After extensive washes with 70% ethanol, the specimens were cleared in 25% N, N, N', N'-Tetrakis-(2-hydroxypropyl) ethylenediamine (Tokyo Chemical Industry) in glycerol overnight and mounted in glycerin jelly. Images were acquired using a light-field microscope and analyzed with ImageJ. The area of lower meibomian glands was quantified using typically 8 to 9 separate glands from the nasal edge of the eyelid, depending on the atrophic involution (gland loss or dropout) of individual test animals.

Immunohistochemistry for human sections

Five- μ m-thick eyelid paraffin sections were antigen-retrieved by pressure cooking in citrate buffer (pH 6.0) as described^{10,32,81} and immersed into PBS containing 0.1% Tween-20 (PBS-T). Sections were blocked with 3% BSA in PBS-T for 2 h and incubated with a specific set of antibodies, including anti-CD38 antibody (rabbit monoclonal, EPR4106, abcam, ab108403, 1:1000 dilution) and anti-HSD3B1 (mouse monoclonal, 3C11-D4, Abnova, final 0.05 μ g/ml)⁸² for 24 h at 4°C. The immunoreactivities were visualized with 3,3-diaminobenzidine using horseradish peroxidase-labeled anti-IgG polymers (Dako, EnVision+ System-HRP Labeled Polymer anti-rabbit for CD38 and anti-mouse for HSD3B1). The number of CD38-positive cells was normalized to the area of interstitial tissue of the tarsal plate captured by microscopy.

Immunofluorescence staining

Cryosections (16 μ m thick) of fresh-frozen eyelids were fixed in 4% paraformaldehyde in PBS at room temperature for 15 min and blocked in PBS containing 3% bovine serum albumin and 0.1% Triton X-100 for 2 h at room temperature. The sections were then stained with a specific primary antibody in PBS containing 0.1% Triton X-100 for 24 h at 4°C. The primary antibodies used include anti-CD38 rabbit monoclonal antibody EPR21079, abcam, ab216343 (1:600 dilution) and anti-CD38 rabbit monoclonal antibody E9F5A, Cell Signaling, 68336 (1:100 dilution). The immunoreactivities were visualized using Alexa488-conjugated anti-rabbit IgG (Thermo Fisher Scientific, 1:1000). The sections were then counterstained with 4',6-diamidino-2-phenylindole (DAPI). Fluorescence images were obtained using a BZ-X710 fluorescence microscope (Keyence). For co-labeling with a macrophage marker, paraffin sections were subjected to immunofluorescence staining using anti-CD38 antibody (rabbit monoclonal, E9F5A, Cell Signaling, 68336, 1:100) and anti-CD11b (rat monoclonal, M1/70, Thermo Fisher Scientific, 14-0112-82, 1:100) and visualized using corresponding secondary anti-bodies: Alexa647-conjugated anti-rabbit IgG and Alexa488-conjugated anti-rat IgG (both from Thermo Fisher Scientific, 1:1000).

Immunoblotting

The tarsal plate samples were lysed into Laemmli buffer containing 1 $\times$ cOmplete Protease Inhibitor cocktail (Roche Diagnostics) as described^{83,84}. Immunoblotting was performed as described^{83,84} using antibodies for CD38 (abcam, ab216343, 1:3000 dilution) and β -Actin (Sigma, A5441, 1:1000).

Transcriptome profiling via microarray and RNA-seq

Transcriptome of *Hsd3b6*^{+/+} and *Hsd3b6*^{-/-} mouse meibomian glands were depicted using DNA microarray by adopting GeneChip Mouse Gene 2.0 ST Array (Thermo Fisher Scientific). The tissue samples analyzed include the laser-microdissected meibomian gland samples from *Hsd3b6*^{+/+} and *Hsd3b6*^{-/-} mice sacrificed at circadian time (CT) 0 and 12, comprising two biologically independent samples per group, each generated from a pool of twelve eyelids. Expression data were subject to background correction and normalization using the RMA algorithm implemented in Affymetrix Expression Console. Pseudogenes, predicted genes, and genes without a symbol or a specific GenBank NM number were excluded. Genes with values higher than the mean plus two standard deviations of the seminal proteins (*Svs3a*, *Svs3b*, *Svs5*, *Svs6*, *Sva*, *Sval1*, and *Sval2*), which are unlikely to be present in the eyelid, were considered expressed. To identify genotype-dependent DEGs under potential circadian variations, expression values were normalized at each CT point based on the wild-type average and genotype-dependent differences were extracted and assessed using the limma (Linear Models for Microarray Data⁸⁵) package with the Benjamini-Hochberg correction for multiple testing. Genes were considered DEGs if their adjusted *P* was < 0.1 and the fold change value was > 1.25 or < 0.8 in microarray (Fig. 1⁸⁵). For studying the transcriptomes of young and aged C57BL/6J wildtype mouse meibomian glands with or without 78c-NMN treatment, we used RNA-seq as described^{10,84}. RNA-seq libraries were prepared using a NEBNext Ultra II Directional RNA Library Prep kit for Illumina (NEB). Libraries were paired-end-sequenced on a NovaSeq 6000 sequencer (Illumina). Sequenced reads were mapped on the mouse genome GRCm39 and quantified using the CLC Genomics Workbench (v24.0.1). Ribosomal RNA genes and pseudogenes were removed from analysis. Differential expression analysis was conducted using DESeq2 (v1.46.0), applying an adjusted *P* value threshold of < 0.1, and MA-plots were generated using the log2-transformed fold change and the mean of normalized read counts for each gene. GO enrichment was calculated with the GOrilla online tool (<http://cbl-gorilla.cs.technion.ac.il/>⁸⁶). The complete list of enriched GO terms is available in Supplementary Data 1 (related to Fig. 1⁸⁶), Supplementary Data 2 (related to Fig. 4⁸⁶), and Supplementary Data 3 (related to Fig. 5⁸⁶).

Quantitative RT-PCR

Total RNA was extracted using a Fast Gene RNA basic kit (Nippon Genetics). cDNA was synthesized using FastGene Scriptase II (Nippon Genetics) and quantitative real-time PCR was performed using THUNDERBIRD SYBR qPCR mix (TOYOBO) and StepOnePlus (Applied Biosystems). *Actb* was used as an internal control. The primer sets used in this study included: *Actb*, FW: 5'-CCA GCC TTC CTT CTT GGG TA-3', REV: 5'- AGA GGT CTT TAC GGA TGT CAA CG-3', *Tnf*, FW: 5'- CTC CCT CTC ATC AGT TCT ATG-3', REV: 5'- TTT GCT ACG ACG TGG GCT AC-3', *Ccl3*, FW: 5'- TGA AAC CAG CAG CCT TTG CTC-3', REV: 5'- AGG CAT TCA GTT CCA GGT CAG TG-3', *Il1b*, FW: 5'- TGC CAC CTT TTG ACA GTG ATG-3', REV: 5'- TGA TGT GCT GCT GCG AGA TT-3', *Il6*, FW: 5'- AGA CAA AGC CAG AGT CCT TCA GAG-3', REV: 5'- AGC ATT GGA AAT TGG GGT AGG AAG G-3', *Cd38*, FW: 5'- CTG GGA CGC TGC CTC ATC TAC-3', REV: 5'- GGG GCG TAG TCT TCT CTT GTG-3', *Sirt1*, FW: 5'- CAG TGT CAT GGT TCC TTT GC-3', REV: 5'- CAC CGA GGA ACT ACC TGA T-3', *Parp1*, FW: 5'- GGC AGC CTG ATG TTG AGG T-3', REV: 5'- GCG TAC TCC GCT AAA AAG TCA C-3', *Nampt*, FW: 5'- GAA CGT GCT GTT CAC AGT G-3', REV: 5'- CAG ACC ATC TAA GTT ACC AG-3', *Nmnat1*, FW: 5'- AGT TAC CCA CAA AGC TCA C-3', REV: 5'- TCC ATC TTC CAC AAG TTG-3', *Nmnat2*, FW: 5'- TGT AGA TGA GAA CGC CAA CC-3', REV: 5'- AGT CCC CAA CAA TCA CTT CC-3', *Nmnat3*, FW: 5'- TGG ATG GAA ACG GTG AAG-3', REV: 5'- TGG AAG GTC TTG AGG ACA TC-3'.

Quantification and statistical analysis

We used ImageJ software for quantification of images, which include quantification of the size of the meibomian glands in whole-mount lipid staining, the number of CD38-positive cells in immunofluorescence, and the intensity of protein bands in immunoblot analysis. All data are expressed as means ± SEM. Statistical tests used are specified in the text and figure legends. Experiments with one variable were analyzed by Student's *t* test or one-way ANOVA using

Bonferroni's multiple comparison test; experiments involving two variables were analyzed by two-way ANOVA with Tukey's multiple comparison test. All statistical analyses were performed with GraphPad Prism 8.

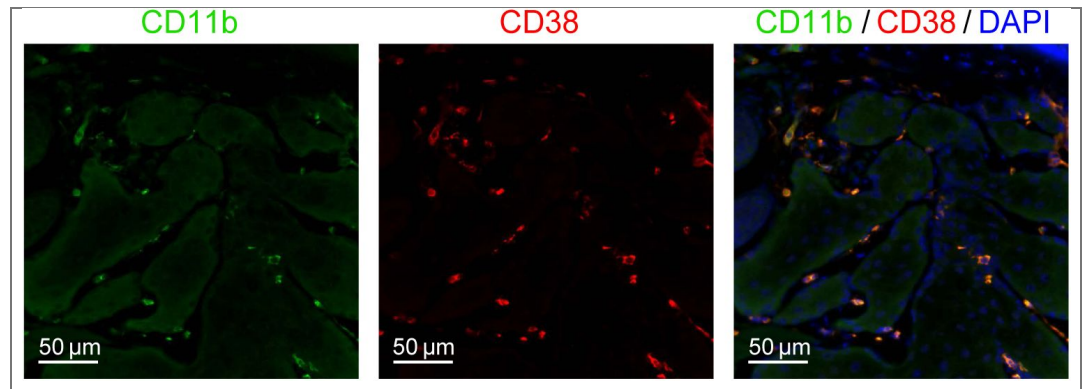


Figure S1. Eyelid CD38 expression and co-labelling with a macrophage marker CD11b, related to Fig. 2. Representative immunofluorescence images of CD38 (red) and a macrophage marker CD11b (green) in *Hsd3b6*^{-/-} eyelid.

Figure S2. Human female eyelid sections, immunolabeled for CD38, related to Fig. 3.

Anti-CD38 and anti-HSD3B1 immunohistochemistry using a pair of serial sections from 85-year-old and 34-year-old female eyelid specimen. The dotted boxes indicate the region of magnified view. CD38-immunopositive cells are pointed by arrowheads.

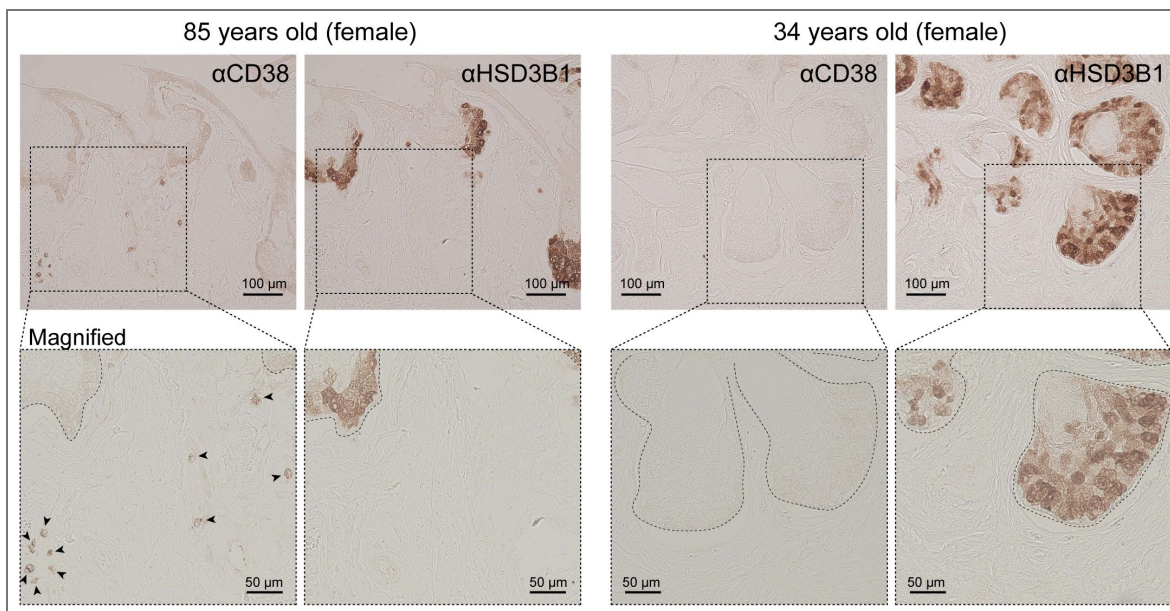
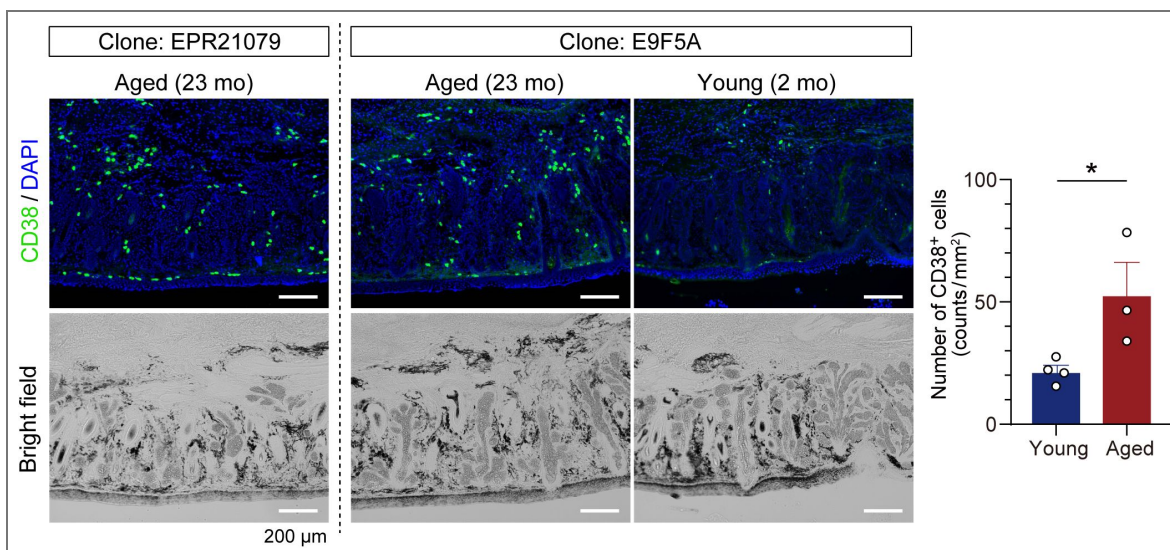


Figure S3. Two different monoclonal antibodies for the mouse CD38 — clones EPR21079 and E9F5A — exhibit similar staining patterns in the tarsal plate of aged mice, related to Fig. 3.

The graph shows the number of CD38 positive cells around meibomian acini in young (2 mo, $n = 4$) and aged (23 mo, $n = 3$) mice when using clone E9F5A. Data are means $\pm$ s.e.m. * $P < 0.05$, unpaired t -test.



Data availability

Microarray and RNA-seq data have been deposited in the Gene Expression Omnibus and are publicly available as of the date of publication. Accession numbers are GSE301740 and GSE302030, respectively.

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Additional information

Author contributions

M.D. conceived the project; M.D. and Y.H. designed the research; Y.H. performed experiments in collaboration with T.S., D.Y., H.U., X.S., T.M., and E.H.; M.D. and Y.H. wrote the paper with input from all authors.

Resource availability

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Masao Doi (doimasao@pharm.kyoto-u.ac.jp [↗](#)).

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Additional files

[Supplementary Data 1](#) [↗](#)

[Supplementary Data 2](#) [↗](#)

[Supplementary Data 3](#) [↗](#)

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Peer reviews

Reviewer #1 (Public review):

Summary:

While the results show some loss in the eyelid meibomian glands, there is significant gland retention in HSD3b6 KO mice, as shown in Figure 2. This is supported by the lack of DEG patterns showing downregulation of Meibum lipid genes (AWAT2, Far2, Soat1, Plin2, SCD, etc.), and no decrease in Pparg expression, known to be critical for meibomian gland lipid gene expression.

Weaknesses:

It should be noted that while the authors indicate that CD38 is significantly up-regulated in the HSD3b6 KO mouse, the increase was not sufficient to show a significant adjusted P-value. Bulk RNA sequencing also shows no significant change in meibum lipid gene expression for aged mice that are treated with 78c, an inhibitor of CD38, which the authors indicate increases NAD levels, leading to increased meibomian gland size compared to vehicle-treated mice. Unfortunately, there was no increase in meibum lipid gene expression with 78c, as identified by adjusted P-value. However, it should be noted that the supplemental file covering DEG expression was labeled as a Microarray analysis. This did not include the 78c+NMN treated mice, which the authors contend show a more impactful effect on the meibomian gland.

<https://doi.org/10.7554/eLife.110696.1.sa3>

Reviewer #2 (Public review):

Summary:

In this manuscript, the authors demonstrate strong correlations between a pro-inflammatory state, the activity of an intracrine hormone (3 beta-hydroxysteroid dehydrogenase, 3B-HSD), and the NAD co-factor. Specifically, in a 3B-HSD knockout mouse, there was an upregulation

in pro-inflammatory cytokines and increased CD38⁺ cells (CD38 is an enzyme that depletes NAD, a necessary cofactor for 3B-HSD activity). Conversely, induction of inflammation in the eyelids resulted in reductions in 3B-HSD activity. Supplementation with 5 alpha-dihydrotestosterone (DHT) or the NAD precursor NMN, and inhibition of CD38 activity (78c), corrected the pathologies observed in both the 3B-HSD knockout mouse and the pro-inflammatory model (LPS injection into eyelids).

Strengths:

The experiments were performed with good rigor, assessing the impact of inflammation and 3B-HSD activity using multiple model systems. The endpoints represented a combination of transcriptional changes, protein quantification, enzymatic activity, and immunofluorescent microscopy. The authors use human tissue from both younger and older individuals to justify their hypotheses that increased CD38⁺ cells and reduced 3B-HSD quantity exist in older individuals. The data provide the foundation for assessing more global changes to the tear film and ocular surface.

Weaknesses:

The main weaknesses of the study include the following:

- (1) An absence of information on meibomian gland health, tear film, and ocular surface.
- (2) Too few human subjects to validate the hypotheses.

Conclusion:

Overall, this study demonstrates an important relationship that exists between intracrine signaling, inflammation, and cofactor signaling. It represents a novel approach in therapeutic design for patients with meibomian gland dysfunction.

<https://doi.org/10.7554/eLife.110696.1.sa2>

Reviewer #3 (Public review):

Summary:

The authors aimed to investigate whether disruption of intracrine steroid hormone metabolism contributes to meibomian gland dysfunction and proposed a "vicious cycle" of gland dysfunction and inflammation, using a global *Had3b6* knockout mouse model. The work addresses an important aspect of MGD, but its impact may be limited unless the intracrine mechanism can be more clearly distinguished from systemic hormonal effects.

Strengths:

This study addressed an important question. The hormonal regulation of the meibomian gland has long been recognized. If clarified, the concept of local steroid metabolism influencing gland homeostasis could have implications for understanding disease mechanisms and identifying therapeutic targets.

Weaknesses:

The use of a global knockout makes it difficult to separate local intracrine effects from systemic hormonal changes, and key controls and hormone measurements are lacking.

LPS-induced inflammation may not reflect the chronic nature of MGD.

<https://doi.org/10.7554/eLife.110696.1.sa1>

Author Response:

Public Reviews:

Reviewer #1 (Public review):

Summary:

While the results show some loss in the eyelid meibomian glands, there is significant gland retention in HSD3b6 KO mice, as shown in Figure 2. This is supported by the lack of DEG patterns showing downregulation of Meibum lipid genes (AWAT2, Far2, Soat1, Plin2, SCD, etc.), and no decrease in Pparg expression, known to be critical for meibomian gland lipid gene expression.

Weaknesses:

It should be noted that while the authors indicate that CD38 is significantly up-regulated in the HSD3b6 KO mouse, the increase was not sufficient to show a significant adjusted P-value. Bulk RNA sequencing also shows no significant change in meibum lipid gene expression for aged mice that are treated with 78c, an inhibitor of CD38, which the authors indicate increases NAD levels, leading to increased meibomian gland size compared to vehicle-treated mice. Unfortunately, there was no increase in meibum lipid gene expression with 78c, as identified by adjusted P-value. However, it should be noted that the supplemental file covering DEG expression was labeled as a Microarray analysis. This did not include the 78c+NMN treated mice, which the authors contend show a more impactful effect on the meibomian gland.

We thank the reviewer for the careful evaluation and insightful comments regarding the interpretation of meibomian gland phenotypes and gene expression profiles.

Regarding the point on the apparent retention of meibomian gland structure and the lack of downregulation of key lipid-related genes (e.g., *Awat2*, *Far2*, *Soat1*, *Plin2*, *Scd*, and *Pparg*), we agree that these observations are important for interpreting the extent of gland dysfunction. In the revised manuscript, we will more clearly present and discuss the RNA-seq data, including the expression profiles of representative meibomian gland lipid genes (and other DEGs), to better contextualize these findings.

With respect to *Cd38* expression, we acknowledge that the statistical significance based on adjusted P-values was limited in the current microarray dataset. To address this point, we will perform additional validation using targeted quantitative PCR with specific primers to more accurately assess *Cd38* expression changes.

Reviewer #2 (Public review):

Summary:

In this manuscript, the authors demonstrate strong correlations between a pro-inflammatory state, the activity of an intracrine hormone (3 beta-hydroxysteroid dehydrogenase, 3B-HSD), and the NAD co-factor. Specifically, in a 3B-HSD knockout mouse, there was an upregulation in pro-inflammatory cytokines and increased CD38+ cells (CD38 is an enzyme that depletes NAD, a necessary cofactor for 3B-HSD activity). Conversely, induction of inflammation in the eyelids resulted in reductions in 3B-HSD activity. Supplementation with 5 alpha-dihydrotestosterone (DHT) or the NAD precursor NMN, and inhibition of CD38 activity (78c), corrected the pathologies observed in both the 3B-HSD knockout mouse and the pro-inflammatory model (LPS injection into eyelids).

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Weaknesses:

The main weaknesses of the study include the following:

- (1) An absence of information on meibomian gland health, tear film, and ocular surface.*
- (2) Too few human subjects to validate the hypotheses.*

Conclusion:

Overall, this study demonstrates an important relationship that exists between intracrine signaling, inflammation, and cofactor signaling. It represents a novel approach in therapeutic design for patients with meibomian gland dysfunction.

We thank the reviewer for the positive evaluation of our study and for recognizing the rigor of the experiments, the use of multiple model systems, and the potential of the data to provide a foundation for further investigation.

Regarding the points raised under weaknesses, we agree that evaluation of meibomian gland function, tear film, and ocular surface phenotypes would provide important additional insight. In the present study, we focused primarily on the structural phenotype of the meibomian gland, particularly gland size, as a primary feature of MGD. We acknowledge that pathological assessments of gland function and ocular surface conditions have not been fully addressed. We will clearly state this limitation and expand the Discussion to position these aspects as important directions for future investigation.

With respect to the limited number of human samples, we acknowledge that this is an important consideration for validating the translational relevance of our findings. We will revise the manuscript to more explicitly address this limitation and interpret the human data with appropriate caution.

Reviewer #3 (Public review):

Summary:

The authors aimed to investigate whether disruption of intracrine steroid hormone metabolism contributes to meibomian gland dysfunction and proposed a "vicious cycle" of gland dysfunction and inflammation, using a global Had3b6 knockout mouse model. The work addresses an important aspect of MGD, but its impact may be limited unless the intracrine mechanism can be more clearly distinguished from systemic hormonal effects.

Strengths:

This study addressed an important question. The hormonal regulation of the meibomian gland has long been recognized. If clarified, the concept of local steroid metabolism influencing gland homeostasis could have implications for understanding disease mechanisms and identifying therapeutic targets.

Weaknesses:

The use of a global knockout makes it difficult to separate local intracrine effects from systemic hormonal changes, and key controls and hormone measurements are lacking.

LPS-induced inflammation may not reflect the chronic nature of MGD.

We thank the reviewer for the thoughtful evaluation and for highlighting the importance of distinguishing intracrine mechanisms from systemic hormonal effects.

We agree that, as currently presented, the use of a global *Hsd3b6* knockout model makes it difficult to fully separate local intracrine effects from systemic hormonal changes. This point is also consistent with the major concern raised in the editorial assessment regarding the need to more clearly establish the proposed intracrine mechanism. To address this issue, we will strengthen the evidence for intracrine regulation by incorporating additional analyses. Specifically, we will assess systemic testosterone levels in *Hsd3b6* knockout mice and include appropriate controls using orchidectomized (ORX) mice. These analyses will help to better distinguish local intracrine mechanisms from systemic hormonal influences.

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